

36118 Applied Natural Language Processing
Master of Data Science and Innovation, UTS

Assessment Task 2B

End-to-End NLP Project — Technical Report

Sydney Liveability Explorer

Subject 36118 Applied Natural Language Processing
Submission AT2 Part B — Group Technical Report
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GitHub Repository <https://github.com/Nelkit/sydney-liveability-ai>

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1 Executive Summary

Write this section last. Include one paragraph summarising each of the following: project objectives, data sources, NLP methods, findings, and outcomes.

2 Project Objectives and Scope

Describe the problem this project solves, the target user group, and what is in and out of scope for this submission.

3 Project Phases and Team Contributions

Describe project phases and map responsibilities to each team member. Each member must include an individual paragraph describing their specific contribution.

Phase Breakdown Table

Phase	Tasks	Team Member(s)
Phase 1: Data	Data acquisition and preprocessing from Reddit (Arc Shift), Community Insights PDF, BOCSAR CSV, TfNSW GTFS, OpenStreetMap (Overpass API), and City of Sydney ArcGIS.	
Phase 2: NLP Pipeline	Aspect-level sentiment analysis (DistilRoBERTa), sentence embeddings (MiniLM), ChromaDB ingestion, spaCy preprocessing.	
Phase 3: Multi-Agent Pipeline	CrewAI pipeline crew (pre-computation) and query crew (real-time), FastAPI backend, RAG with Claude API.	
Phase 4: Frontend	Next.js dashboard, Leaflet map, aspect radar chart, emotion profile visualisation, conversational chat UI.	
Phase 5: Evaluation and Report	NLP metrics, RAG quality, application evaluation, report, poster, peer review.	

Juan

This team member contributed to the data ingestion and retrieval layers of the pipeline, with a focus on making community-sourced qualitative data accessible to the language model.

The primary contribution was the design and implementation of the full ingestion pipeline for the City of Sydney community consultation PDFs. Starting from structured JSON output produced by the PDF parser, each text chunk was embedded using the all-MiniLM-L6-v2 sentence transformer model and written to the ChromaDB collection `sydney_liveability` with metadata including suburb, theme, source document, page number, and chunk index. This pipeline processed 1,114 records across three official PDF reports, forming the qualitative backbone of the RAG system. A

secondary contribution was the completion of the Synthesiser agent's ChromaDB retrieval layer, which embeds each user query at runtime, retrieves the top five semantically relevant chunks, and injects them into the LLM prompt alongside structured PostgreSQL outputs and specialist agent results, enabling the model to cite resident quotes and PDF extracts by source and suburb.

4 Data Sources

4.1 Primary Data Sources

Document each source: name, access method, format, granularity, and contribution to the pipeline. Sources include: Community Insights PDF, Reddit via Arc Shift, BOCSAR CSV, TfNSW GTFS, OpenStreetMap via Overpass API, and City of Sydney ArcGIS.

Community Insights PDF

This source is an official qualitative data source comprising three PDF documents published by the City of Sydney: the *Community Insights Report 2024*, the *Haymarket Vision Engagement Report*, and the *Haymarket Summary*. These reports capture structured resident feedback gathered through formal consultation processes, covering themes such as transport, housing, public spaces, environment, culture, safety, and social inclusion across suburbs including Haymarket, Sydney, Redfern, Belmore, Ultimo, Burwood, and Camperdown.

Raw PDFs were processed using a custom Python parser that extracted and segmented text into thematic chunks, tagging each record with its source document, suburb, theme, and page number. The pipeline produced four JSON files containing a total of 1,114 records across the three source documents. These structured records were subsequently ingested into ChromaDB as sentence embeddings, enabling the RAG pipeline to retrieve suburb-specific qualitative evidence in response to user queries.

This source provides a dimension of liveability insight that quantitative datasets cannot capture: the lived experience and stated priorities of Sydney residents as documented by their local government. Its official provenance and thematic breadth make it a high-quality complement to the sentiment data collected from Reddit.

4.2 Data Limitations and Ethical Considerations

Discuss Reddit as a public but self-selected source, privacy considerations, geographic bias toward inner Sydney suburbs, PDF granularity limitations, and the ethical use of community-generated content.

Community Insights PDF

This source presents limitations that should be considered when interpreting pipeline outputs.

Text was extracted from structured reports rather than raw survey responses, meaning some granularity is lost in the aggregation process. The reports summarise resident feedback through an editorial lens applied by the City of Sydney, which may omit minority opinions or nuanced dissenting views that did not align with the overall consultation narrative.

Geographic coverage is uneven. The majority of records are concentrated in Haymarket (383 records) and Sydney CBD (108 records), while suburbs such as Camperdown and Burwood contribute only a handful of entries. This imbalance means the RAG pipeline is better equipped to answer queries about inner-city areas than about less-represented suburbs.

Suburb attribution relied on named entity recognition and heuristic matching during PDF parsing, which introduces a small margin of error in cases where suburb names are ambiguous or absent from the source text.

From an ethical standpoint, all three PDFs are publicly published by the City of Sydney and were used solely for research purposes, with no modification to the original content. The community feedback they contain was gathered through formal, consented consultation processes. However, using officially summarised community views to draw conclusions about neighbourhood liveability carries a responsibility to avoid misrepresenting resident priorities or presenting aggregated findings as exhaustive.

5 NLP Methods and Techniques

5.1 Text Preprocessing

Describe the preprocessing pipeline using spaCy: tokenisation, lemmatisation, and named entity recognition (NER) for suburb extraction. Also describe LangChain chunking strategy used for RAG ingestion.

5.2 Considered Approaches: Traditional NLP

Present a critical review of traditional NLP methods (VADER, LDA, TF-IDF) with reference to the literature. Explain why these approaches were considered but not adopted as the primary implementation for this project. Do not present them as implemented components.

5.3 Implemented Approaches: Modern NLP

5.3.1 Reddit Data Collection via Arc Shift

Explain why PRAW was replaced by Arc Shift for data collection. Report scale: 20,237 posts covering 563 suburbs.

5.3.2 Aspect-Level Sentiment Analysis (DistilRoBERTa)

Describe the eight sentiment dimensions analysed per suburb, the emotion profiling approach, and implementation by Kai (Ying-Kai Liao).

5.3.3 Sentence Embeddings (all-MiniLM-L6-v2) and ChromaDB

Describe the chunking strategy, embedding model, ChromaDB upsert process, and metadata fields (suburb, source) stored per chunk.

5.3.4 Multi-Agent Pipeline (CrewAI)

Describe both crews: the pipeline crew (pre-computes suburb data) and the query crew (responds in real time). Detail each agent, its role, tools, and how agents collaborate.

5.3.5 Retrieval-Augmented Generation (RAG + LLM)

Detail the RAG chain built with LangChain and ChromaDB. Describe how the Claude API (Sonnet) generates answers grounded in retrieved source chunks with citations.

5.4 Justification: Why Modern Over Traditional

Present an academic argument for each design choice: DistilRoBERTa over VADER, sentence embeddings over TF-IDF, RAG over keyword search, and CrewAI for modular orchestration. Cite Lewis et al. (2020), Reimers & Gurevych (2019), and Hutto & Gilbert (2014).

6 Findings and Evaluation

6.1 NLP Pipeline Results

Summarise aspect scores per suburb, emotion profiles, transport scores, POI density, and crime trends. Reference Appendix D for full tables and charts.

6.2 NLP Model Performance Metrics

Report evaluation metrics for DistilRoBERTa, retrieval quality for the RAG pipeline, and representative query-response pairs with citations from the LLM.

6.3 User-Facing Application Quality

Describe the dashboard, aspect radar visualisation, emotion profile chart, chat interface, and overall user experience quality based on walkthroughs or testing.

6.4 Temporal Trends

Summarise time-based patterns in Reddit sentiment per suburb where timestamps are available.

7 Outcomes and Value Added

Describe what the project achieved and the practical value delivered to the target user group.

8 Recommendations and Next Steps

List the highest-impact extensions: coverage of more suburbs, additional data sources, model fine-tuning, and integration with rental price APIs.

9 Conclusion

Summarise the project contributions, lessons learned, and future potential.

10 References

Use APA 7th edition for all citations. Key references include: DistilRoBERTa (Sanh et al., 2019), all-MiniLM-L6-v2 (Reimers & Gurevych, 2019), RAG (Lewis et al., 2020), CrewAI, and Arc Shift.

A Appendix A: System Architecture Diagram

Insert the full pipeline diagram: Arc Shift, GTFS, OSM, BOCSAR, and PDF ingestion feeding into DistilRoBERTa and ChromaDB, then CrewAI, RAG, and the frontend.

B Appendix B: Dashboard Screenshots

Insert annotated screenshots of the aspect radar, emotion profile, Leaflet map, and chat interface.

C Appendix C: Sample Code

Include representative code for `ingest_sentiment.py`, CrewAI agents, the RAG chain, and FastAPI endpoints.

D Appendix D: Evaluation Tables and Charts

Include suburb-level scores per sentiment dimension, emotion distributions, transport scores, and POI density charts.

E Appendix E: Traditional vs. Modern NLP Comparison Table

Table 2: Traditional vs. Modern NLP Approach Comparison

Approach	Type	Used	Rationale
VADER sentiment	Traditional	No	Lexicon-based, no aspect granularity
LDA topic modelling	Traditional	No	Unsupervised, low coherence on Reddit text
TF-IDF keyword extraction	Traditional	No	Frequency-based, no semantic understanding
DistilRoBERTa aspect sentiment	Modern	Yes	8-dimension aspect scoring, contextual
all-MiniLM-L6-v2 embeddings	Modern	Yes	Semantic similarity, compact and fast
RAG with Claude Sonnet	Modern	Yes	Grounded, cited answers from retrieved chunks
CrewAI multi-agent pipeline	Modern	Yes	Modular orchestration, scalable per suburb